

YT Content Creator — Project Review

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Executive Summary

Syntax errors: **None** (all files pass `py_compile`)
Import errors: **None** when `PYTHONPATH=project root`
Critical issues: **2** | High: **4** | Medium: **8** | Low: **6**
Missing from requirements.txt: **pydub**
ffmpeg in nixpacks.toml: **Yes**
PYTHONPATH in start cmd: **Correct (/app)**

1. app/core/config.py

- L8 — ValueError if `MAX_VIDEO_DURATION_MINUTES` env is non-numeric
- L8-10 — Defaults OK; overridable via env
- Missing — `YOUTUBE_COOKIES` not in config (used in downloader via `os.environ`)

2. app/core/database.py

- L37-45 — `save_video()` new UUID on INSERT OR REPLACE orphans `generated_content`
- L75-82 — `get_content_for_video()` keeps OLDEST duplicate format, not newest
- L28 — No ON DELETE CASCADE on FK (`delete_video` handles manually)

3. app/core/downloader.py

- L46-55 — Hardcoded timeout, retries, User-Agent
- L74,86 — duration null causes TypeError on division
- L84 — `get_video_info()` + download each create cookie temp files
- L137-141 — Error mentions sidebar cookies upload but UI has none
- L117-120 — Requires ffmpeg (nixpacks OK)

4. app/core/transcriber.py

- L3,11 — `GROQ_API_KEY` bound at import from config
- L19-20,44 — Hardcoded 24MB / 10min chunk limits
- L39-41 — pydub MISSING from requirements.txt
- L43 — pydub requires ffmpeg

5. app/core/content_engine.py

- L3,19 — Import-time `GROQ_API_KEY`
- L31,38-43 — Hardcoded temperature, truncation, `max_tokens`
- Prompt files — All 4 exist in prompts/

6. app/exporters/exporter.py

- L14 — Unused import `TA_JUSTIFY`
- L52 — Unused import `Pt`
- L37-44 — Incomplete XML escape in PDF headings

7. app/pages/02_history.py

- L22 — duration_sec None causes TypeError on // 60
- L22-27 — Nested expanders may warn in Streamlit
- L57-71 — No try/except on PDF/DOCX export (unlike main.py)

8. app/main.py

- L3-11 — All core imports correct
- L38-55 — api-badge/hero-badge CSS removed; cosmetic
- L147-149 — XSS risk: transcript in unsafe_allow_html
- L234-235 — Copy button does not copy to clipboard
- L100,151 — None transcript crashes on .split()

9. requirements.txt

- MISSING: pydub (used for large audio files)
- Has: streamlit, yt-dlp, groq, python-dotenv, reportlab, python-docx
- Note: Full pip-freeze style (58 pkgs), not minimal pins

10. railway.toml

- L5 — PYTHONPATH=/app correct
- L5 — streamlit run app/main.py correct
- L6 — healthcheckPath=/ may fail with Streamlit startup

11. nixpacks.toml

- L2 — ffmpeg included
- L2 — python312 may duplicate auto-detected Python
- L5 — PYTHONPATH=/app correct; matches railway.toml

Cross-Cutting Checks

main.py imports: **All resolve correctly**

requirements vs code: **pydub missing**

Local dev: set PYTHONPATH to project root before streamlit run

Priority Fixes

1. Add pydub to requirements.txt
2. Fix get_content_for_video() — keep newest per format (database.py:82)
3. Fix save_video() UUID replace orphaning content (database.py:37-45)
4. Escape transcript HTML (main.py:147-149)
5. Fix bot error message — YOUTUBE_COOKIES not sidebar (downloader.py:140)
6. Guard duration_sec None (downloader.py:86, 02_history.py:22)
7. Fix Copy button or use st.clipboard (main.py:234)
8. Add export try/except in 02_history.py

Issues Table

Severity	File	Line	Issue
Critical	requirements.txt	—	pydub missing
Critical	main.py	147-149	XSS via unsafe HTML transcript

High	database.py	82	Wrong cached content version
High	database.py	37-45	Orphaned content on re-save
High	downloader.py	140	Misleading cookies error message
High	transcriber.py	39	pydub not in requirements
Medium	main.py	234	Copy button misleading
Medium	02_history.py	22	None duration_sec crash
Medium	downloader.py	86	None duration crash
Low	exporter.py	14,52	Unused imports